CS 306: Media Access Control

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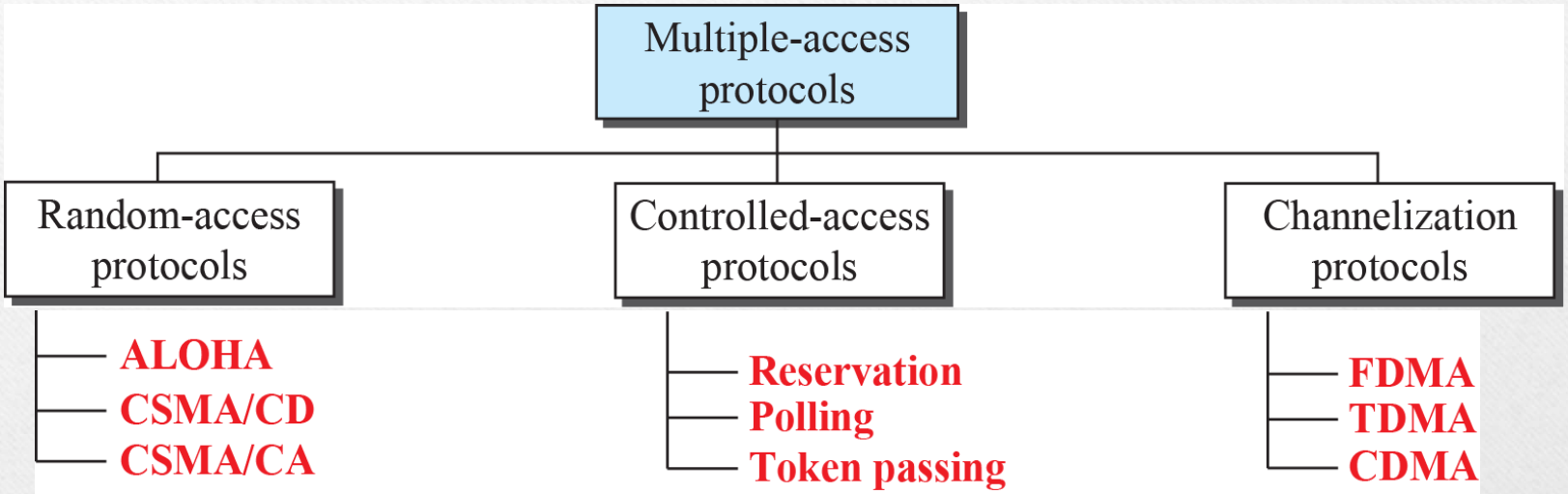
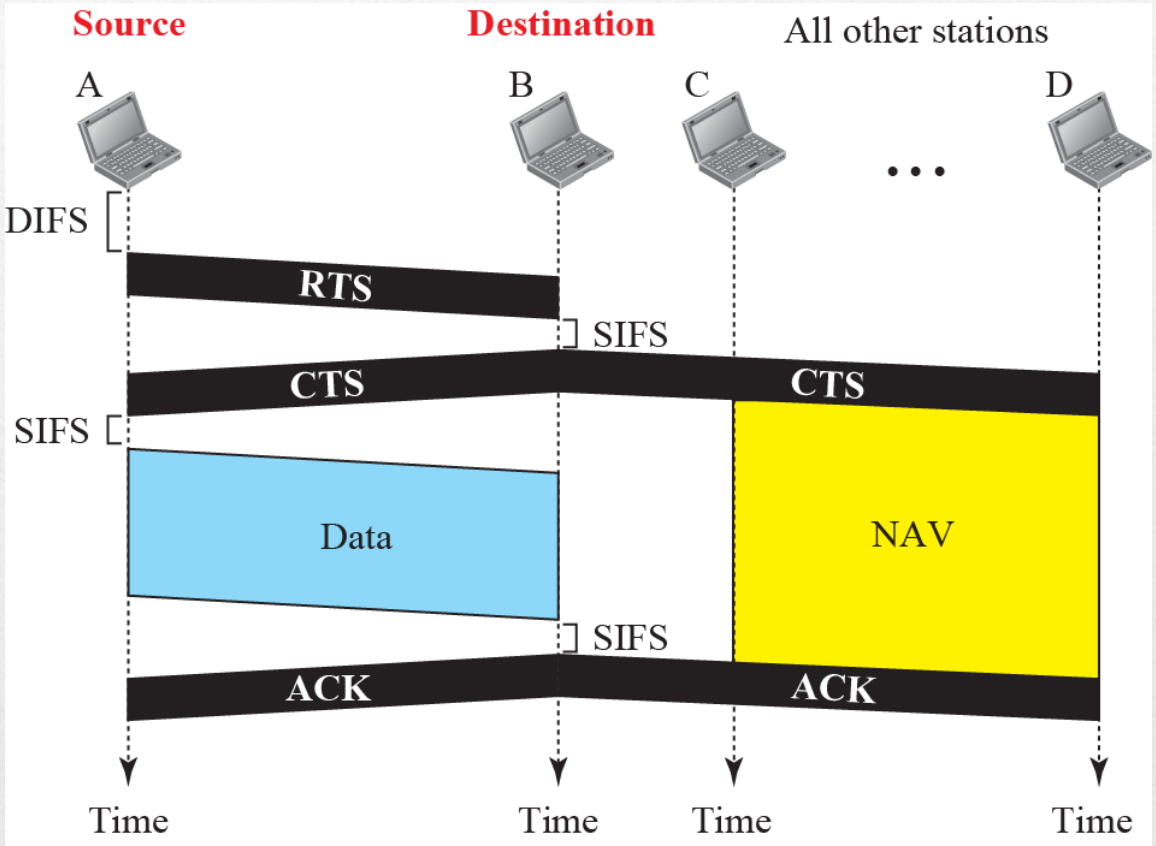
Outline

- CSMA/CA
- Controlled Access
- Channelization

Taxonomy of multiple-access protocols



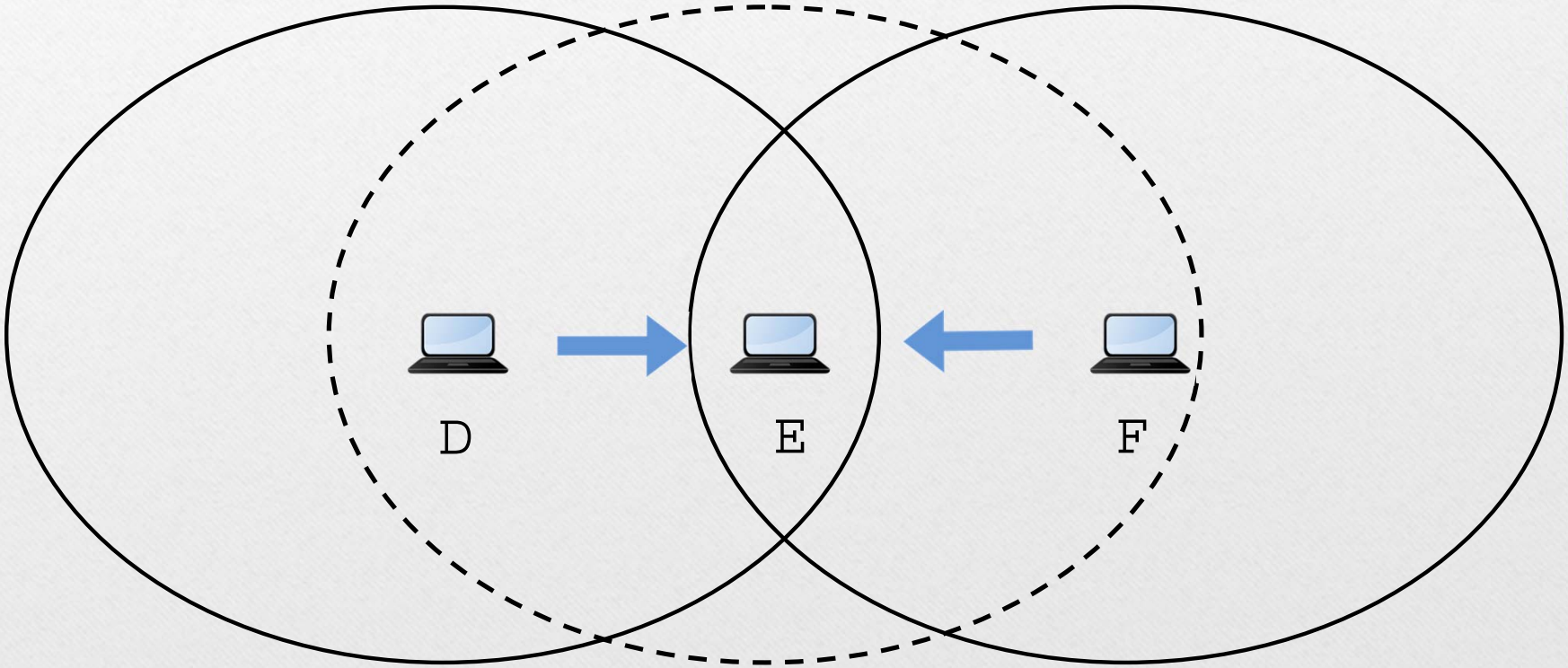
CSMA/CA


Source
Destination
All other stations
A
B
C
D
DIFS
RTS
SIFS
CTS
CTS
SIFS
Data
NAV
SIFS
ACK
ACK
Time
Time
Time
Time

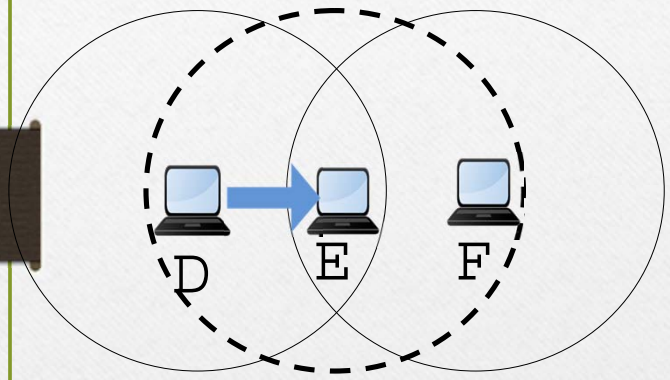
CSMA/CA (Contd.)

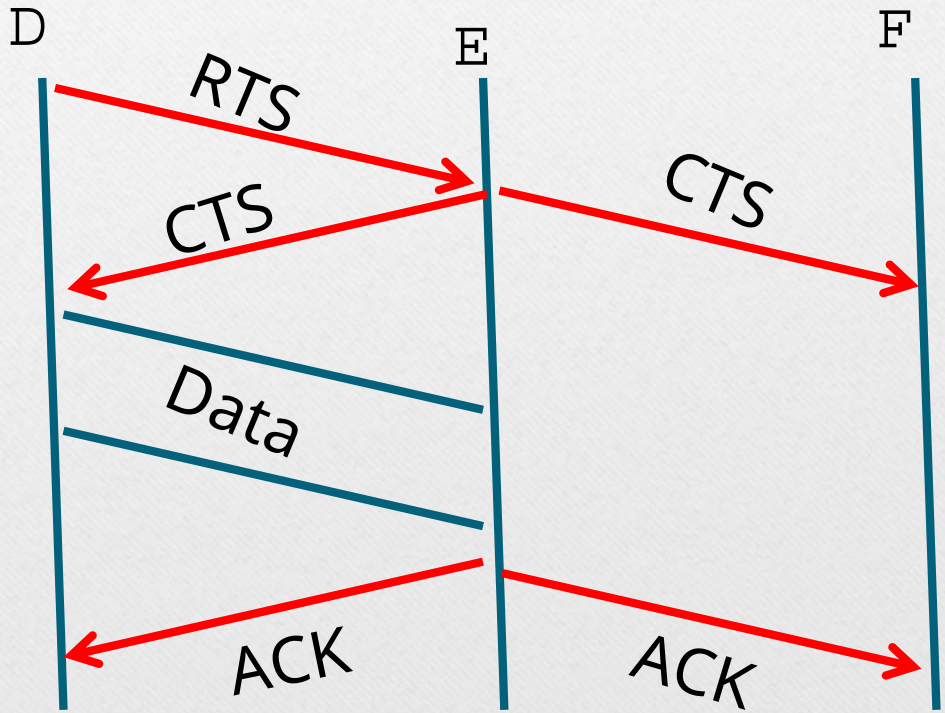
- Request to Send (RTS)
- Clear to Send (CTS)
- Network Allocation Vector (NAV)
- Acknowledgement (ACK)

Hidden Terminal Problem

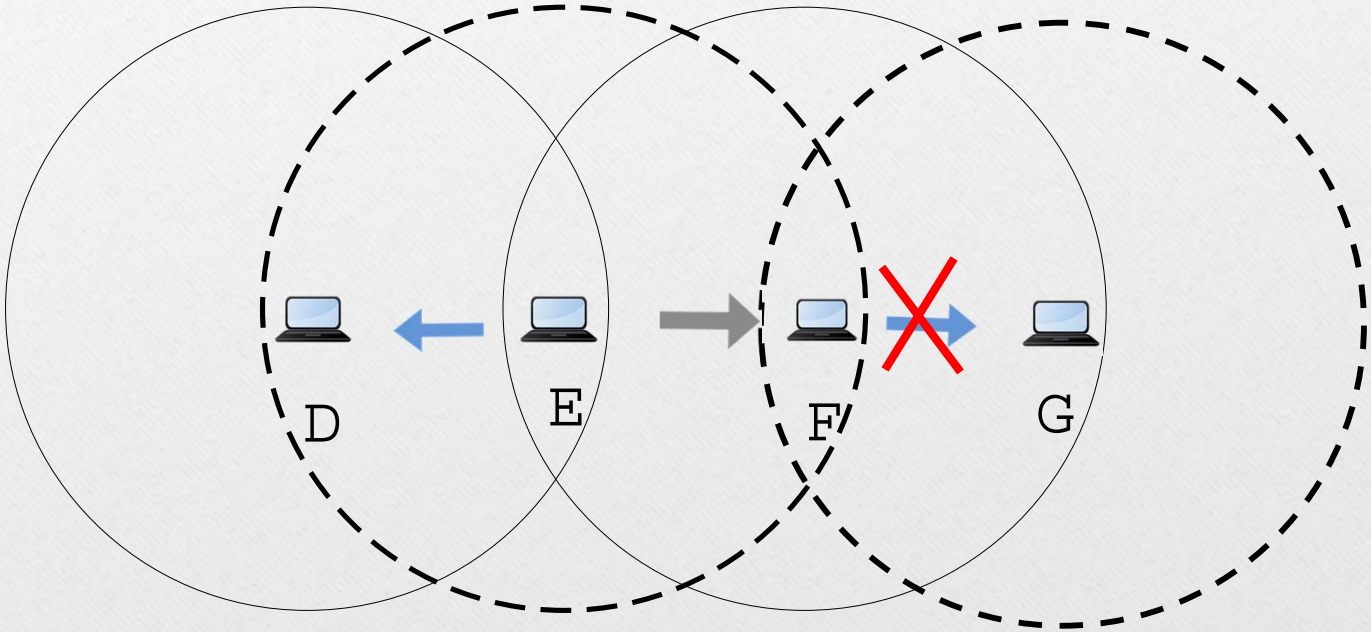


Hidden Terminal Solution

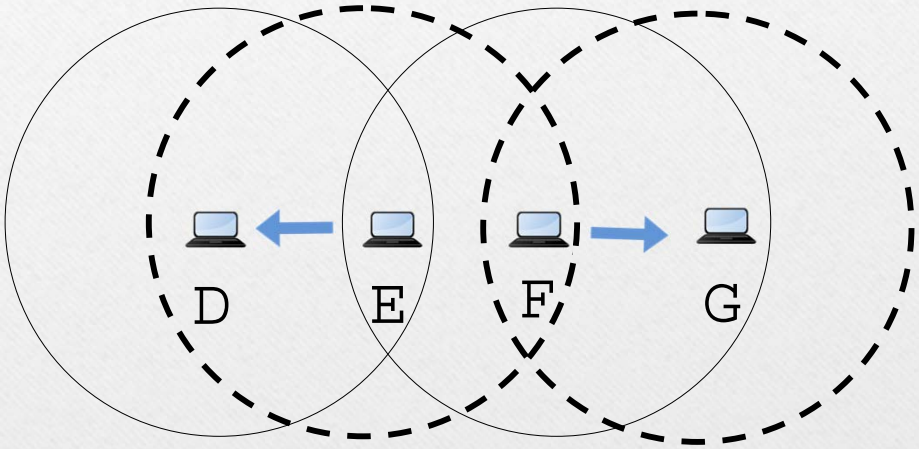


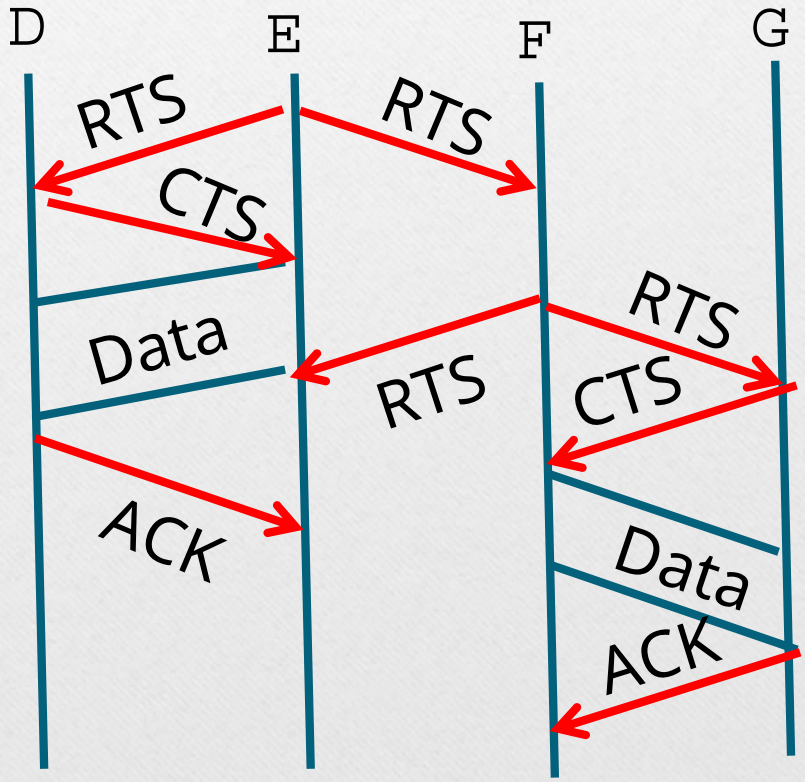


Exposed Terminal Problem



Exposed Terminal Solution





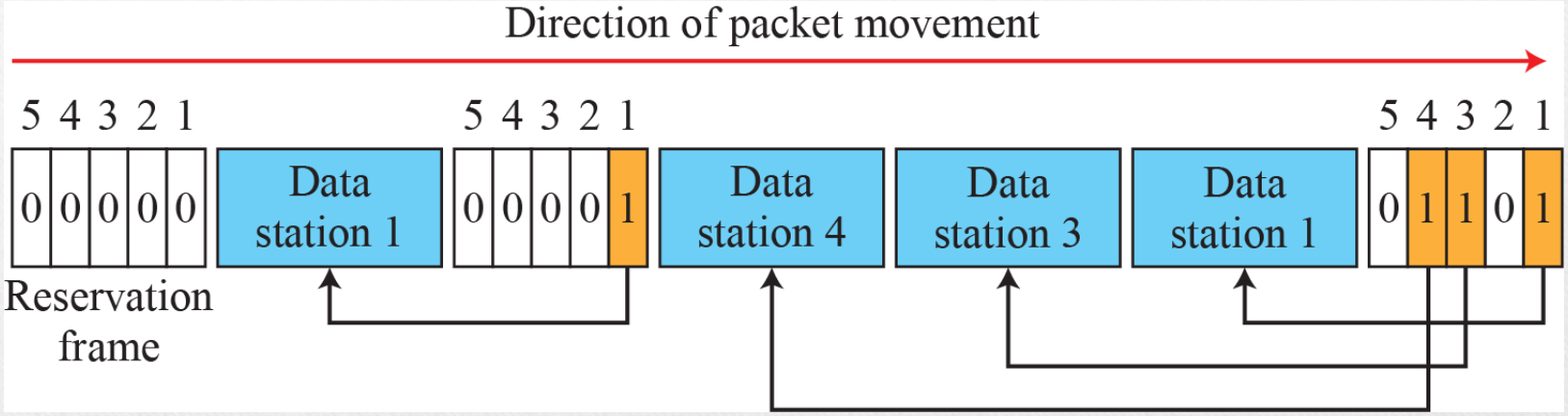
Controlled Access

- In controlled access, the stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.

Reservation

- In the reservation method, a station needs to make a reservation before sending data.
- Time is divided into intervals.
- In each interval, a reservation frame precedes the data frames sent in that interval.

Reservation access method

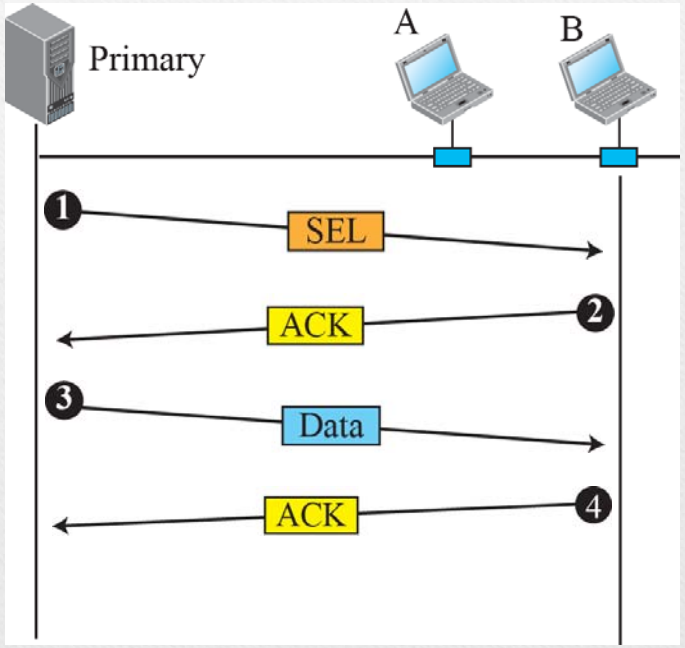


Polling

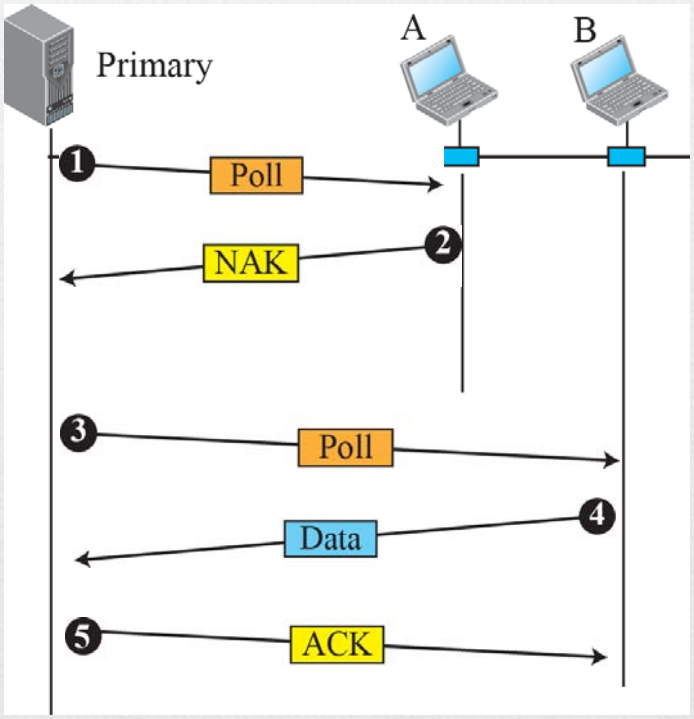
- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations.
- All data exchanges must be made through the primary device even when the ultimate destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions.

Select and poll functions in polling-access method

Select



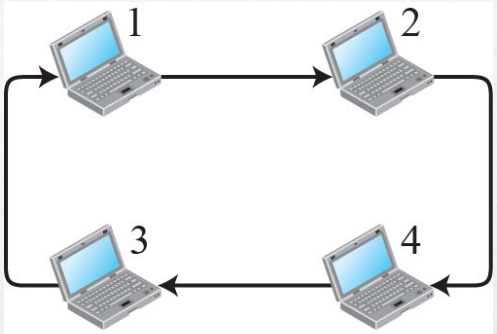
Poll



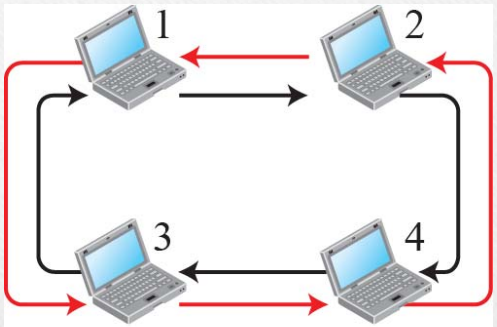
Token Passing

- In the token-passing method, the stations in a network are organized in a logical ring.
- In other words, for each station, there is a predecessor and a successor.
- The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring.

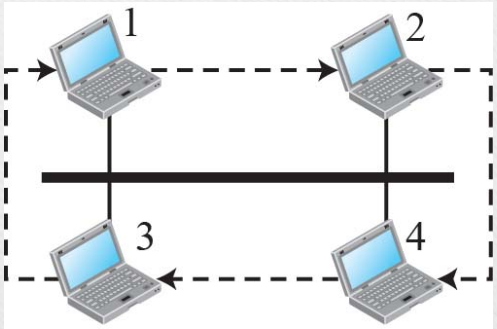
Token-passing access method



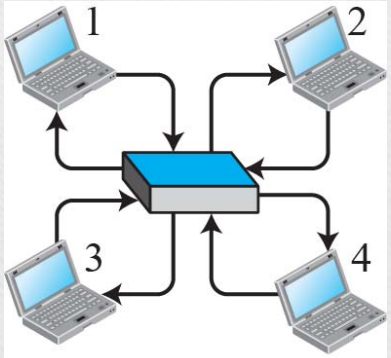
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

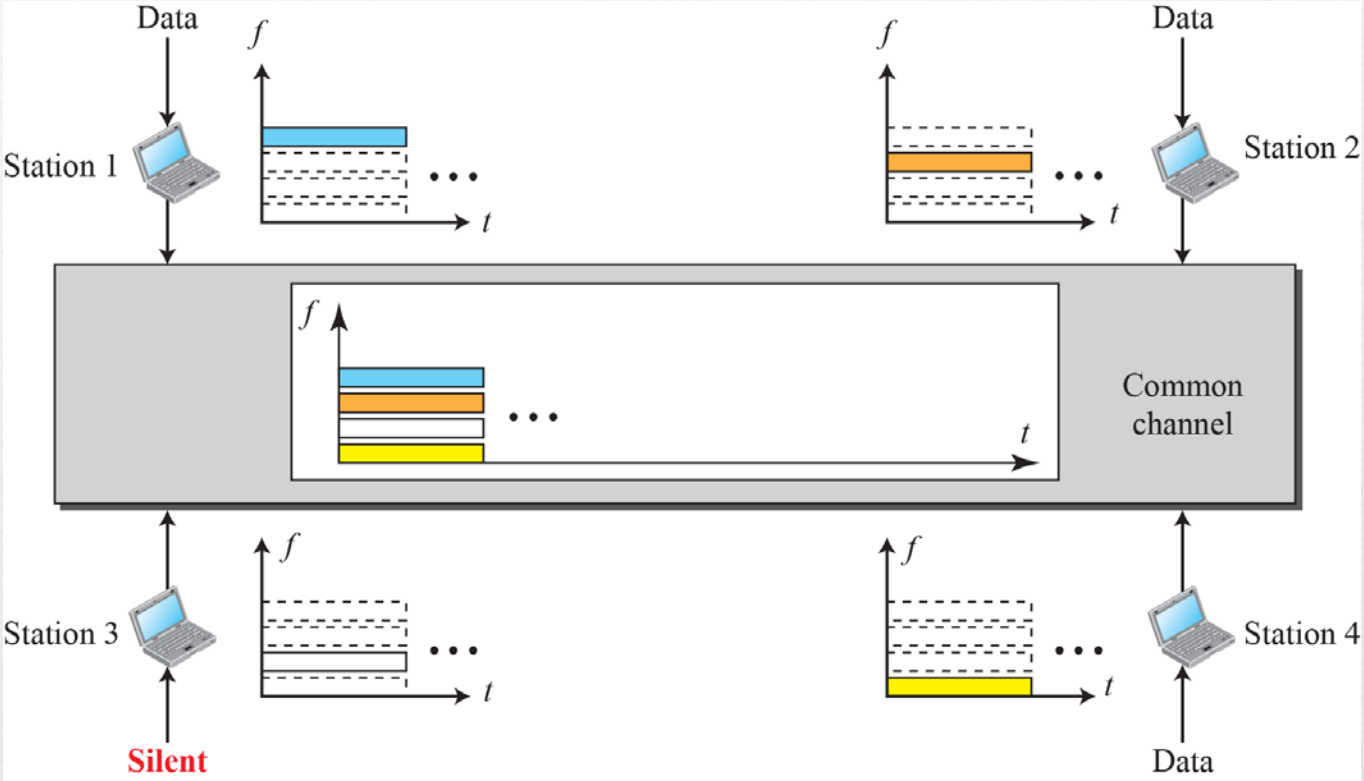
Channelization

- Channelization (or channel partition, as it is sometimes called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations.
- In this section, we discuss three protocols:
  - FDMA, TDMA, and CDMA.

Frequency-Division Multiple Access

- In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands. Each station is allocated a band to send its data.
- In other words, each band is reserved for a specific station, and it belongs to the station all the time.
- Each station also uses a bandpass filter to confine the transmitter frequencies.

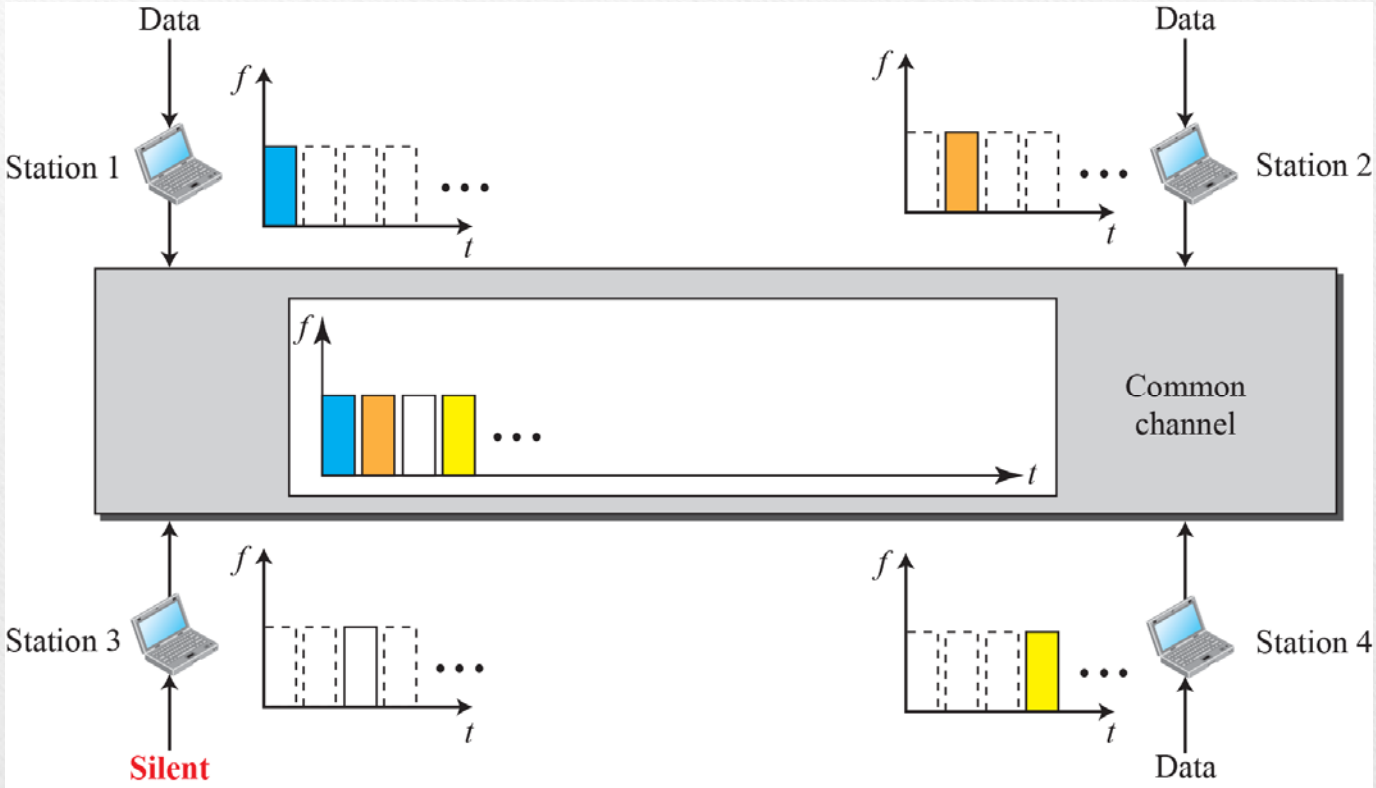
FDMA (Contd.)


Data
Station 1
Data
Station 2
Common channel
Station 3
Silent
Station 4
Data

Time-Division Multiple Access

- In time-division multiple access (TDMA), the stations share the bandwidth of the channel in time.
- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot.

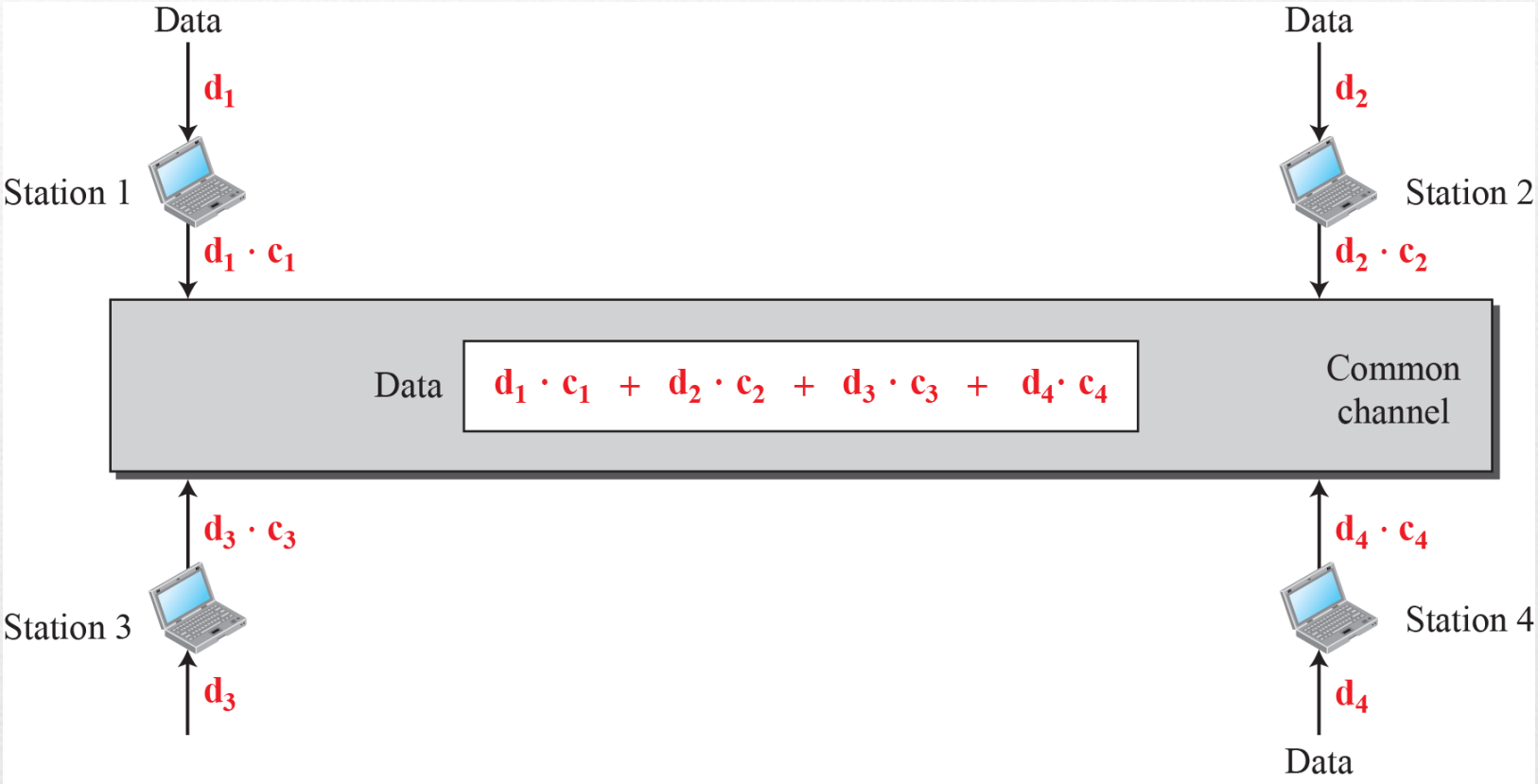
TDMA (Contd.)


Data
Station 1
f
t
Data
Station 2
f
t
f
t
Common
channel
Station 3
f
t
Silent
Station 4
f
t
Data

Code-Division Multiple Access

- Code-division multiple access (CDMA) was conceived several decades ago. Recent advances in electronic technology have finally made its implementation possible.
- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link.
- It differs from TDMA in that all stations can send data simultaneously; there is no timesharing.

Simple idea of communication with code



Chip sequences

C_1

$[+1 \quad +1 \quad +1 \quad +1]$

C_2

$[+1 \quad -1 \quad +1 \quad -1]$

C_3

$[+1 \quad +1 \quad -1 \quad -1]$

C_4

$[+1 \quad -1 \quad -1 \quad +1]$

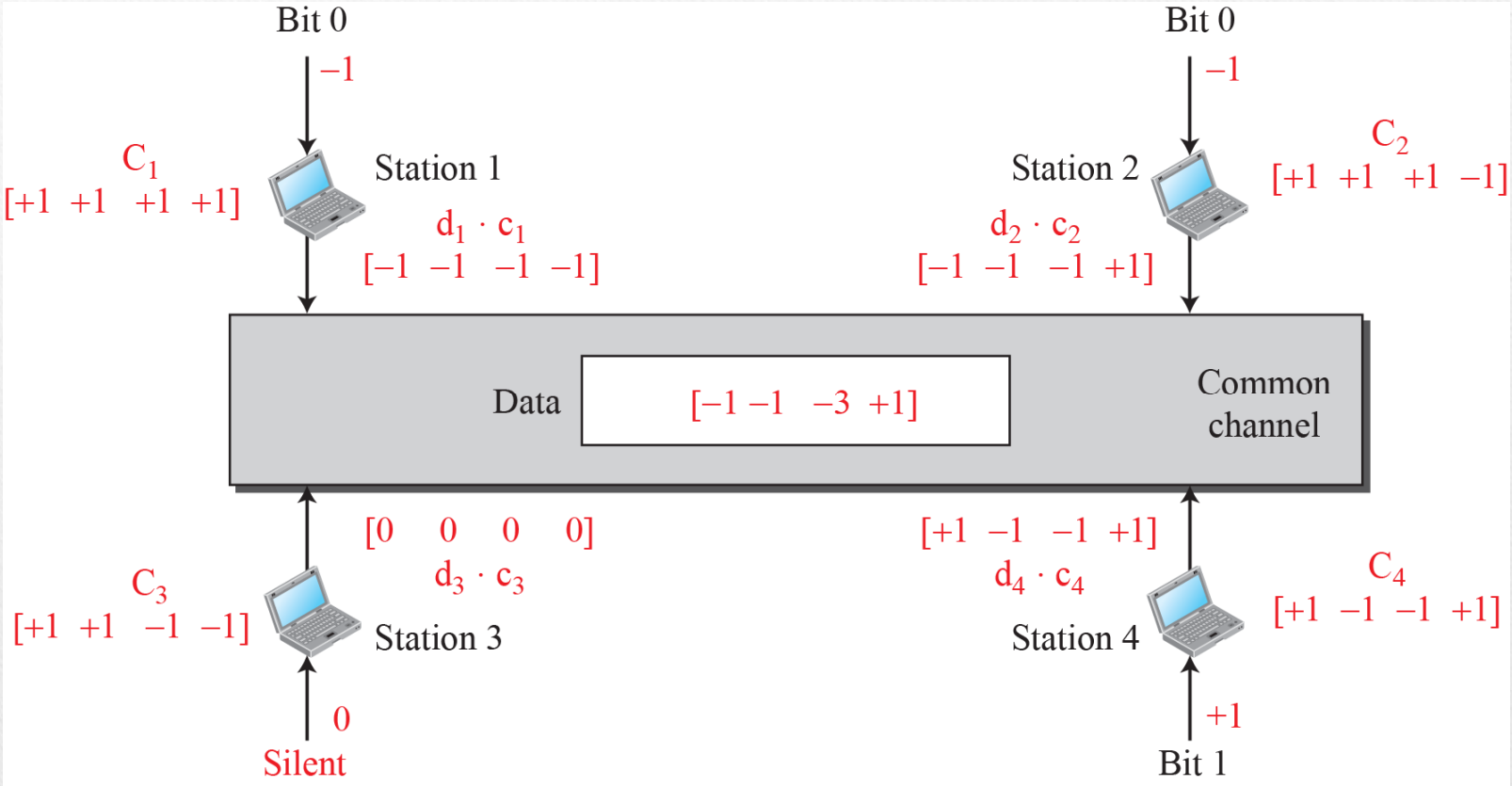
Data representation in CDMA

Data bit 0 → −1

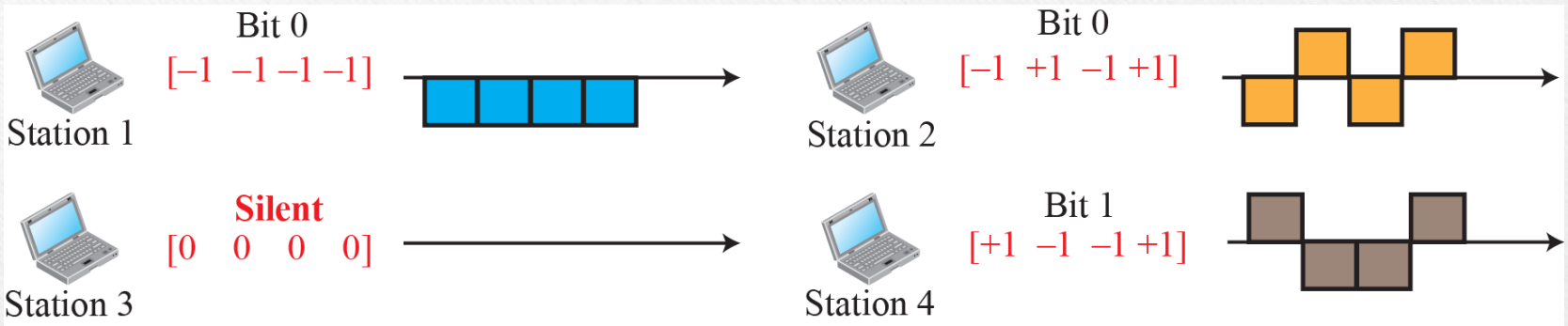
Data bit 1 → +1

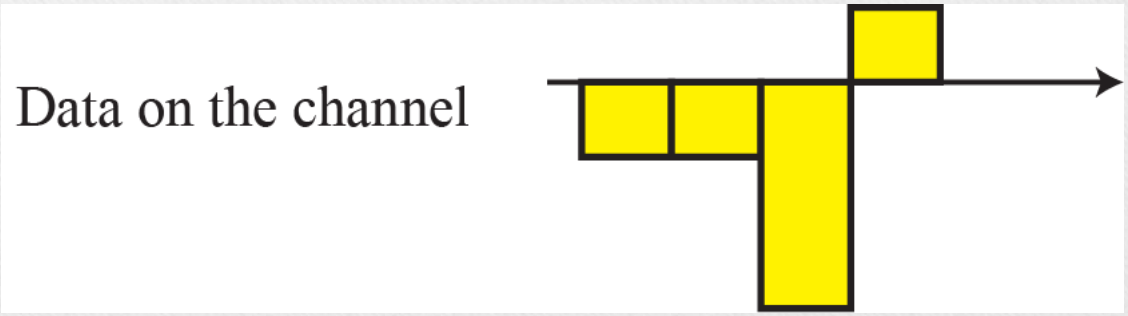
Silence → 0

Sharing channel in CDMA

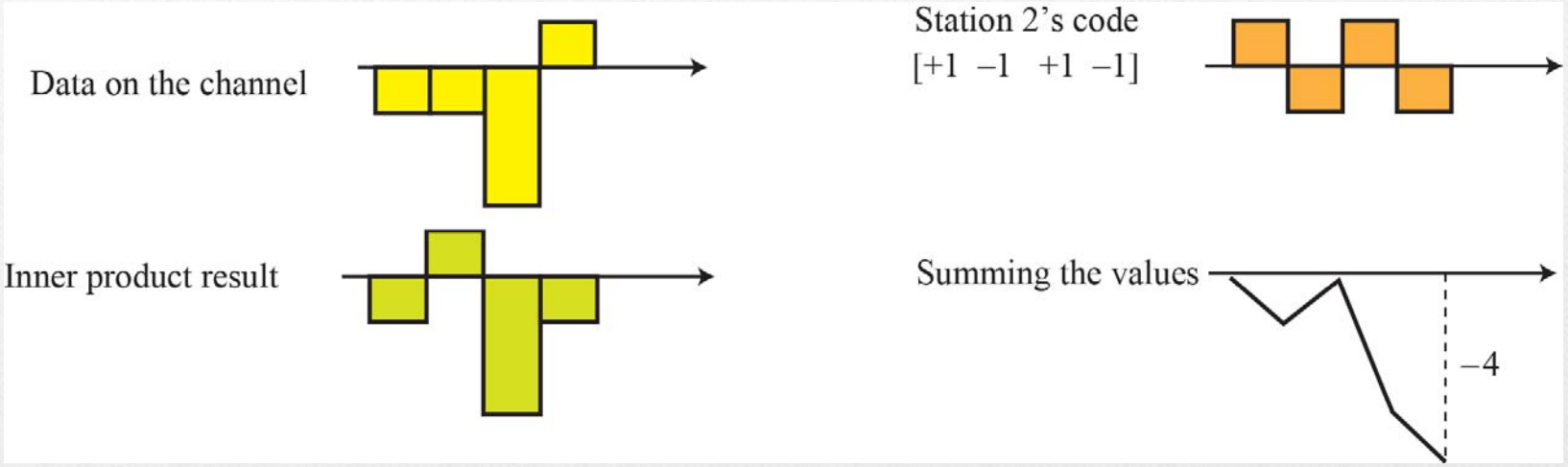


Digital signal created by four stations in CDMA





Decoding of the composite signal for one in CDMA



–4 → –4/4 → –1 → Bit 0

Example

- What is the number of sequences if we have 90 stations in our network?

Solution

The number of sequences needs to be 2^m. We need to choose $m = 7$ and $N = 2^7$ or 128. We can then use 90 of the sequences as the chips.

Example

- Find the chips for a network with
  - Two stations
  - Four stations

Solution

We can use the rows of W_2 and W_4 in Figure 6.34:

a. For a two-station network, we have [+1 +1] and [+1 −1].

b. For a four-station network we have [+1 +1 +1 +1], [+1 −1 +1 −1], [+1 +1 −1 −1], and [+1 −1 −1 +1].

Example

- Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.

Solution

Let us prove this for the first station, using our previous four-station example. We can say that the data on the channel $D = (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4)$. The receiver, which wants to get the data sent by station 1, multiplies these data by c_1.

$$
\begin{aligned}
[D \cdot c_1] / 4 &= [(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1] / 4 \\
&= [d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1] / 4 \\
&= [d_1 \times 4 + d_2 \times 0 + d_3 \times 0 + d_4 \times 0] / 4 = [d_1 \times 4] / 4 = d_1
\end{aligned}
$$

Next Class

- Wired Network

?

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